

Series and infinite products – 2

How fast the terms must decay: Abel, Dirichlet, Cesàro

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Theoretical minimum.

Theorem 1 (Abel summation and the two tests). With $B_n = b_1 + \dots + b_n$,

$$\sum_{n=1}^N a_n b_n = a_N B_N - \sum_{n=1}^{N-1} (a_{n+1} - a_n) B_n.$$

Hence Dirichlet's test: if $a_n \downarrow 0$ and the partial sums B_n are bounded, then $\sum a_n b_n$ converges; and Abel's test: if (a_n) is monotone and bounded and $\sum b_n$ converges, then $\sum a_n b_n$ converges. The inequality behind these is developed in Inequalities – 6; here summation by parts is used as a convergence tool.

Theorem 2 (Olivier). If $a_n \geq 0$ is non-increasing and $\sum a_n$ converges, then $na_n \rightarrow 0$. Monotonicity is essential: without it a convergent series may have $na_n \not\rightarrow 0$, and even $\limsup na_n = \infty$.

Theorem 3 (Kronecker's lemma). If $\sum \frac{a_n}{n}$ converges, then $\frac{a_1 + \dots + a_n}{n} \rightarrow 0$. Conversely, bounded Cesàro means force convergence of $\sum \frac{a_n}{n^{1+\varepsilon}}$ for every $\varepsilon > 0$, but not for $\varepsilon = 0$. Both directions are one application of Abel summation.

Fact 1 (Abel–Dini: there is no boundary). Let $\sum a_n$ diverge, $a_n > 0$, and $s_n = a_1 + \dots + a_n$. Then $\sum \frac{a_n}{s_n}$ still diverges while $\sum \frac{a_n}{s_n^{1+\varepsilon}}$ converges for every $\varepsilon > 0$; symmetrically, for a convergent series with remainders r_n the series $\sum \frac{a_n}{r_n}$ diverges. So no divergent series is slowest and no convergent series is fastest.

Fact 2 (the Cauchy criterion in blocks). Convergence is exactly the statement that all long blocks $a_{m+1} + \dots + a_n$ are small. Consequently

statements about sums of blocks of a fixed length are statements about the sequence itself — with an arithmetic obstruction, since blocks of lengths p and q control everything only through $\gcd(p, q)$.

1. Using the Dirichlet and Abel tests (they were proved from Abel summation in Inequalities – 6), determine for which α the series $\sum \frac{\sin n}{n^\alpha}$ and $\sum \frac{\cos n}{n^\alpha}$ converge and for which of them the convergence is absolute; then prove that $\sum \frac{|\sin n|}{n}$ diverges and that $\sum \frac{\sin n}{n^\alpha}$ is not absolutely convergent for any $\alpha \leq 1$.
2. Prove Olivier's theorem, and then construct a convergent series with positive terms for which $\limsup na_n = \infty$; deduce that the monotonicity in Olivier's theorem cannot be replaced by any condition on the order of growth of (a_n) alone. (The Abel–Dini statement of Fact 1 was proved in Inequalities – 6.)
3. (VJIMC 2006) Suppose that $(a_n)_{n \geq 1}$ is a sequence of real numbers such that the series $\sum \frac{a_n}{n}$ is convergent. Show that the sequence $b_n = \frac{1}{n} \sum_{j=1}^n a_j$ is convergent and find its limit.
4. (МФТИ 2007) Let $(a_n)_{n \geq 1}$ be a sequence of real numbers such that the sequence of arithmetic means $b_n = \frac{1}{n} \sum_{k=1}^n a_k$ converges. Prove that for any $\varepsilon > 0$ the series $\sum \frac{a_n}{n^{1+\varepsilon}}$ is convergent.
5. (МФТИ 2016) Let $(x_n)_{n \geq 0}$ be a sequence of real numbers such that the sequences

$$y_n = \sum_{i=0}^{2016} x_{n+i} \quad \text{and} \quad z_n = \sum_{i=0}^{128} x_{n+i}$$

converge. Is it true that the sequence (x_n) converges?